MSE 480 Lab 2

Turning Process Simulation using Fusion 360

Team Members:

- Thanh Nam Tran - 301540228
- Devon Gill - 301378782
- Igor Montecinos Menacho - 301538376

Group Number: 12

Instructor: Dr. Ahad Armin

1. Introduction

This report details the turning procedure performed on a 1.5” diameter 6061 aluminum rod using Fusion 360. The objective is to identify machining parameters, differentiate between roughing and finishing, and evaluate the limitations of machining. A Fusion 360 simulation of the turning process is also included.

---

2. Fusion 360 Drawing

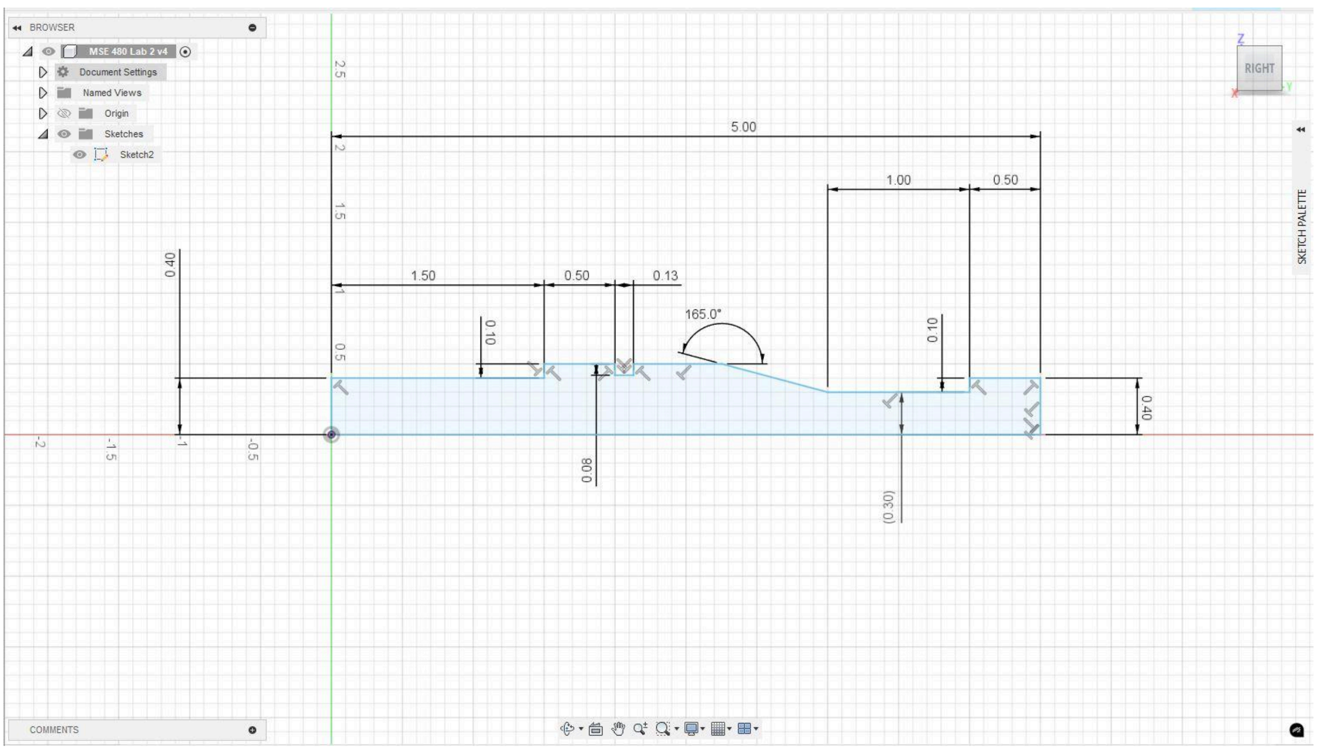

Figure 1: Fusion 360 Drawing of the Part

To effectively make the part on Fusion 360, it must be drawn as shown in Figure 1. The part is cut in half to ensure the revolve command can be used. After the revolve command is executed the part will then be produced adequately as shown in Figure 2.

Figure 2: The Part in Fusion 360 after a Hole is Cut and Executing the Revolve Command

---

3. Turning Procedure

3.1 Material Selection

- **Material Used:** 6061 Aluminum rod
- **Initial Stock Diameter:** 1.5 inches
- **Length of Workpiece:** 5 inches
- **Machining Setup:** Lathe machine with HSS and Carbide tools

3.2 Roughing and Finishing Conditions

3.2.1 Roughing Process

- **Tool Used:** HSS Cutting Tool
- **Cutting Speed:** 250 - 350 SFM (Surface Feet per Minute)
- **Feed Rate:** 0.005 - 0.01 in/rev
- **Depth of Cut:** 0.05 - 0.1 inches per pass
- **Objective:** Remove bulk material efficiently while minimizing tool wear.

3.2.2 Finishing Process

- **Tool Used:** Carbide Cutting Tool
- **Cutting Speed:** 400 - 600 SFM

- **Feed Rate:** 0.002 - 0.005 in/rev
- **Depth of Cut:** 0.01 - 0.02 inches per pass
- **Objective:** Achieve dimensional accuracy and improve surface finish.

3.3 Optimizing Productivity

- **Minimum Waste:** Using precise cutting parameters to avoid unnecessary material removal.
- **Time Optimization:** Using an aggressive but controlled roughing strategy followed by minimal finishing passes.
- **Maximizing Accuracy:** Using fine feed rates and small depth of cuts in finishing to maintain tolerances.

---

4. Surface Finishing Optimization

To achieve a surface roughness of **Ra 32 µin**, the following techniques were used:

- **Use of Carbide Tooling:** Provides a sharper edge, reducing chatter and improving surface quality.
- **Increased Cutting Speed:** Reduces built-up edge formation, leading to a smoother finish.
- **Low Feed Rate (0.002 - 0.005 in/rev):** Ensures gradual material removal for a fine surface.
- **Lubrication/Coolant Use:** Reduces friction, minimizes heat buildup, and improves finish quality.

---

5. Comparison with Additive Manufacturing

5.1 Limitations of Additive Manufacturing in Producing Similar Parts

Aspect	Traditional Machining	Additive Manufacturing
Material Selection	Wide range, including metals	Limited material options
Directional Strength	Isotropic (uniform in all directions)	Anisotropic (weaker in Z-axis)
Surface Finish	Smooth, high precision	Rougher, requires post-processing

Production Time	Fast for single parts	Longer due to layer-by-layer printing
Dimensional Accuracy	High, within tight tolerances	Lower, requires additional machining

- **Material Strength Consideration:** 3D-printed metal parts (e.g., SLM or DMLS) may have weaker mechanical properties due to internal porosity and layer bonding inconsistencies.
- **Time Considerations:** Machining is faster for single parts, while additive manufacturing is beneficial for complex geometries and low-volume production.

---

6. Conclusion

This report covered the complete turning process of a 6061 aluminum rod using Fusion 360. The optimized roughing and finishing procedures ensured minimal waste and high surface quality. A comparison with additive manufacturing highlighted the strengths and limitations of each method. The Fusion 360 simulation effectively demonstrated the machining process.